

The Static Factorization

Theorem:

$$\mathbf{S} \cap \mathbf{C} = \{\mathbf{R1h} = \mathbf{0}\} \cup \{\mathbf{T0h} = \mathbf{0}\} \cup \{\mathbf{T0h} = \mathbf{T0h*}\}$$

$UV_{\xi 3} = 2 \cdot R1h \cdot Mdef_{\xi 5}$ identically · the resting system on the chain = two equations

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Abstract

This monograph records the structural theorem of the static sector of the direct solution programme for the Choptuik problem. It is established that the system of resting equations of the truncated tower, restricted to the static chain of the center, is equivalent to two equations (the UV book, the mass-defect book), while all other residues, including the C1 group, vanish identically, and the books themselves are proportional: $UV_{\xi 3} = 2 \cdot R_{1h} \cdot M_{def_5}$. The branch factorization gives $UV_{\xi 3} = -(16/27)R_{1h}^3 T_{0h}^2 (4T_{0h}^2 - 27)$ and $M_{def_5} = -(8/27)R_{1h}^2 T_{0h}^2 (4T_{0h}^2 - 27)$.

As a consequence, the intersection of the static set with the chain is described exactly: $S \cap C = \{R_{1h} = 0\} \cup \{T_{0h} = 0\} \cup \{T_{0h} = T_{0h}^*\}$ — the trivial family, the half-tower and the flat line; the non-degenerate branch is unique. The numerical census (2501 cells, 56 seeds, 300 starts off the chain, a one-dimensional static kernel) confirms the exhaustiveness: zero new branches. Tables of roots, corollaries for the uniqueness of the clock and the march theorem, honest caveats and the reproduction protocol are included.

Keywords: static factorization, static set, chain, half-tower, flat line, exhaustiveness, SymPy, machine proof.

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1. Introduction

The third monograph records the structural theorem of the static sector: the factorization theorem of the statics on the chain. Obtained in campaign v18b (GLOBAL-STATIC-SEARCH, machine `global_static_search.py`, LEDGER §29) as the exact fact [G1a] and independently confirmed by the v17 contour (machine `clock_closure_t1c.py`, layer C1), it describes the intersection of the static set S with the static chain of the center as an exhaustive union of three families and reduces the whole system of resting equations on the chain to two equations with a common clock structure.

The value of the theorem is twofold. First, it gives a complete classification of the static configurations of the tower on the chain: the trivial family, the half-tower and the flat line with the critical point; there are no other static branches on the chain. Second, it explains the uniqueness of the clock (monograph No. 1): the pair of books is proportional, so two clock conditions never yield two independent scales. The classification is closed beyond the chain as well: a scan of 2501 cells, 56 Gauss–Newton seeds, 300 free-Newton starts off the chain and a static kernel of dimension one (layer G3) find no new non-degenerate branch.

2. Notation and Setting

The static set S is the set of configurations of the truncated tower on which all twelve resting equations vanish. The chain C is the one-parameter family with the exact chain forms $R3h = 2R1h \cdot T0h^2/9$, $W2h = 4T0h^2/3$, $P2h = T0h/3$, $R5h = 2R1h \cdot T0h^2(3T0h^2 - 5)/45$, $M5h = 2R1h \cdot T0h^2(23 - 6T0h^2)/9$ (monograph No. 1, §2.1). The residues $UV_{\xi 3}$ and $Mdef_{\xi 5}$ are the third UV residue and the fifth mass-defect residue; $C1_{\xi 1 \dots \xi 3}$ are the chain constraint group of the first loop.

The setting: to prove that the system of resting equations restricted to C is equivalent to two equations with the same zero set, to write out the factorization of both, and to establish the exact formula for $S \cap C$. Then — to close the question of non-degenerate branches beyond the chain by a numerical census (layers G1–G3).

Table 3.1. The known static families (known_families of machine v18b)

Family	Definition	Status
S0	trivial: all amplitudes 0; $T0h \rightarrow 0$ at any $R1h$	degenerate
S1	the critical point x^* ($T0h = 3\sqrt{3}/2$, $R1h = 1$)	a single point
S2	the flat line: $R1h$ free at $T0h = T0h^*$	the only non-degenerate branch
S3	the half-tower: $T0h$ arbitrary at $R1h = 0$	an exact degenerate branch

3. Statement of the Theorem

Theorem 3 (static factorization on the chain). Let C be the static chain of the center and S the static set of the tower. Then:

(i) all resting equations except two vanish on C identically: the residues $C1_ξ1$, $C1_ξ2$, $C1_ξ3$ are zero as polynomials, and the resting system on C is equivalent to the pair $(UV_ξ3, Mdef_ξ5)$;

$$UV_ξ3 = -(16/27) \cdot R1h^3 \cdot T0h^2 \cdot (4T0h^2 - 27), \quad Mdef_ξ5 = -(8/27) \cdot R1h^2 \cdot T0h^2 \cdot (4T0h^2 - 27) \quad (3.1)$$

(ii) on C the books are proportional identically: $UV_ξ3 = 2 \cdot R1h \cdot Mdef_ξ5$ (ratio_is_2R1h = true, a machine-verified identity); the clock factors of the two books differ by the constant multiple 2, their zero sets coincide;

(iii) the intersection of the static set with the chain is described exactly:

$$S \cap C = \{R1h = 0\} \cup \{T0h = 0\} \cup \{T0h = T0h^*\} \quad (3.2)$$

where $T0h^* = 3\sqrt{3}/2$ is the unique positive root of the clock factor $4T0h^2 - 27$ (the baseline point); on the brick $\delta_C = \pi/7$ the formula survives with the root $T0h = 2.6660683$;

(iv) the numerical census confirms the exhaustiveness: a scan of 2501 cells with 56 seeds finds roots only of the classes flat_line_S2 and trivial_S0; 300 free-Newton starts off the chain give 34 captures of the flat line, ~165 trivial drifts and 0 new branches; the static

kernel dF/da at x^* is one-dimensional (the tangent of the flat line), and the branching attempts over rank drops 4–7 yield no landings off the line.

4. Sketch of the Machine Proof

4.1. The symbolic layer

The proof of clauses (i)–(iii) is executed by SymPy in exact arithmetic. Substitution of the chain forms into the residues yields polynomials in $(R1h, T0h)$; the machine verifies the identical vanishing of the C1 group (residual 0), computes the ratio $UV_{\xi 3}/Mdef_{\xi 5}$ and simplifies it to $2R1h$, then factorizes each book. At the baseline the factorization gives (3.1); on the brick $\delta_C = \pi/7$ both books acquire the common spinor denominator $64827 = 3^4 \cdot 267$ and the cofactors $(\pi - 14)(\pi + 14)$, after which the clock factor takes the form $(196 - \pi^2)T0h^2 - 1323$. The positive root is verified in 50-digit arithmetic: 2.5980762113... and 2.6660683050... respectively.

4.2. The numerical census (layers G1–G3)

Layer G1 is a two-dimensional scan of the residual system $(Mdef_{\xi 5}, UV_{\xi 3})$ over a grid of 2501 cells in the $(R1h, T0h)$ plane with 56 Gauss–Newton seeds: all converged roots belong to the classes `flat_line_S2` ($T0h = T0h^*$, $R1h$ arbitrary — a line) or `trivial_S0` ($T0h \leq 10^{-4}$ at arbitrary $R1h$ — a degenerate trivial branch); `grid_G_min = 0` is attained on the known families. Layer G2 — 300 free-Newton starts off the chain: 34 captures of the flat line, about 165 trivial drifts, zero new branches. Layer G3 — the kernel of dF/da at x^* : rank 13 of 14, the kernel is one-dimensional and coincides with the tangent of the flat line; the branching tests over four to seven rank drops yield no landing off the line. Jointly G1–G3 exhaust the static structure in the neighbourhood of the critical scale.

Table 3.2. The numerical census of the statics (machine `global_static_search.py`)

Layer	Protocol	Outcome
G1	scan of 2501 cells + 56 Gauss–Newton seeds	all roots $\in \{\text{flat_line_S2}, \text{trivial_S0}\}$; 0 new
G2	300 free-Newton starts off the chain	34 line captures, ~165 trivial drifts, 0 new

Layer	Protocol	Outcome
G3	kernel of dF/da at $x^* +$ branching tests (4–7 rank drops)	kernel 1-dimensional (line tangent); no landings off the line

5. Numerical Results

Table 3.3. Representative roots of the G1 scan (baseline $\kappa = 2$)

T0h	R1h	Class	F_max (full system)
2.598076211353316	0.005011872	flat_line_S2	1.39×10^{-17}
2.598076211353316	1.584893192	flat_line_S2	4.26×10^{-13}
2.598076211353316	10.000000000	flat_line_S2	1.46×10^{-11}
2.598076211353316	1.000000000	flat_line_S2	5.68×10^{-14}
9.48×10^{-7}	1.385715109	trivial_S0	3.83×10^{-11}
5.56×10^{-9}	47.480799208	trivial_S0	5.29×10^{-11}

The table illustrates both families: the line S2 with the fixed height T0h* and the free ruler (F from 10^{-14} to 10^{-11} across the range of R1h) and the trivial branch S0 with the height escaping to zero as R1h grows. On the brick $\delta_C = \pi/7$ the picture is the same with T0h* = 2.6660683; the maximum F over all landings remains at the machine zero of the representation.

6. Corollaries

Corollary 3.1 (the only non-degenerate branch). Among the three families of the intersection (3.2) only the flat line is non-degenerate: the half-tower kills the ring clock (amplitudes $\rightarrow 0$ as $R1h \rightarrow 0$, monograph No. 1), and the trivial branch is degenerate by construction. The whole non-degenerate statics of the tower on the chain is the line of equilibria.

Corollary 3.2 (one clock equation). The proportionality $UV_{\xi 3} = 2R1h \cdot Mdef_{\xi 5}$ means that the clock pair defines one equation in two branches — the machine reason for the uniqueness of the root 27/4. The independence of the books is impossible already at the algebraic level, which closes the path to “two clocks” and makes the naive adiabatic incompatibility 9/4 versus 9/16 an artifact (monograph No. 1, Corollary 1.1).

Corollary 3.3 (exhaustiveness around x^*). The layers G1–G3 transfer the classification from the chain to the neighbourhood of the critical point: outside the known families there are no non-degenerate static branches, the linearization kernel is one-dimensional and tangent to the line. This is the foundation on which the DAE march (monograph No. 6) states its claim about the unique solution manifold through x^* .

7. Honest Caveats

The theorem describes the intersection of S with the chain exactly and the neighbourhood of x^* — numerically exhaustively; global components of S far from the critical scale were not sought by shooting or `nsolve` within `v18b` (the programme was transferred to the following campaigns; the honest caveat of ledger `v15–v18`). The numerical layers G1–G3 depend on the grid resolution and initializations: 2501 cells and 300 starts cover the explored region but do not replace a symbolic proof of global uniqueness, which for a nonlinear system off the chain does not exist. Puiseux branches at the soft pole 45/8 (monograph No. 8) may create additional chain representations; the factorization (3.1) refers to the T1c source form. Finally, the classification is carried out for the truncated tower with the derived `dd-law`; a tower enhancement (the O7 programme) may refine the list of families, but the structural identities (i)–(ii) are stable, since they rest on chain forms reproducible at all levels.

8. Reproducibility

Table 3.4. Reproduction protocol for Theorem 3

Step	Command	Result
1	<code>python3 global_static_search.py</code>	<code>results/global_static_search.json</code> ([G1a], G1–G3)

Step	Command	Result
2	python3 clock_closure_t1c.py	results/clock_closure_t1c.json (independent [C1])
3	python3 march_dae_nonlinear.py	results/ march_dae_nonlinear.json (tangent kernel, T2)
4	python3 -m pytest tests/ -q	58 passed (the factorization tests included)

The data correspond to commit 5b256cd; an autonomous copy of the machine `global_static_search.py` with a private copy of the results lives in `einstein_direct/open9/machines/` and runs from its own folder.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/global_static_search.py`; `einstein_direct/results/global_static_search.json` (campaign v18b, theorem [G1a], LEDGER §29).
- [2] Ibid.: `einstein_direct/clock_closure_t1c.py`; `einstein_direct/results/clock_closure_t1c.json` (independent contour [C1], campaign v17).
- [3] Ibid.: `einstein_direct/dae_nonlinear_core.py`; `einstein_direct/results/march_dae_nonlinear.json` (campaign v15, statics S).
- [4] Ibid.: `einstein_direct/README.md` §§1.3, 3.5; `einstein_direct/README_EN.md` §29.
- [5] Monographs No. 1, No. 6, No. 9 of the theorems12 cycle.
- [6] M. D. Roberts. *Gen. Relativ. Gravit.* 21, 907 (1989) — the exact solution on which the solver is verified.